

Project 01 — HTML Fundamentals

The Structural Skeleton of the Web

Summary: This document defines the subject matter for Project 01: HTML5 document anatomy, text hierarchy, media embedding, lists, structured data tables, forms and validation, and semantic layout architecture.

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Chapter I: Instructions

- These exercises are carefully arranged in order of difficulty, from easiest to hardest. We will not consider a successfully completed harder exercise if an easier one is not fully functional.
- Ensure that you have the appropriate permissions and correct directory structure for all your files.
- You must follow the exact submission procedures and naming conventions for every exercise.
- Your exercises will be checked and graded by your fellow classmates during peer evaluations.
- Additionally, your exercises may be validated by automated evaluation scripts. The evaluation is rigorous and strict. To avoid unpleasant surprises, be as thorough as possible.
- You must not leave any extraneous files in your directories other than those explicitly specified in the subject.
- Have a question? Ask the peer on your right. If not, try the peer on your left.
- Your reference guides are official documentation (MDN, man, Git docs) and verified technical resources.
- Examine the examples and edge cases carefully. They contain critical details.

— *Did you check with your right-side neighbor?* —

Chapter II: AI Instructions

The Web Piscine is foundational. Your primary mission is to build solid problem-solving reflexes, genuine technical autonomy, and collaborative peer learning.

In the modern tech ecosystem, generative AI shortcuts are tempting. However, shortcuts prevent the formation of neural connections required for deep engineering intuition.

Key Rules on AI Usage:

- Apply your own reasoning to each exercise before consulting external AI tools.
- Never ask an AI to generate solutions directly for you. You must write every line of code yourself.
- During peer evaluation and exams, you will have to explain every architectural decision and command. If you cannot explain your code, you will receive zero points.
- Good practice: You are confused about a concept like Flexbox or Git staging. You discuss it with your peer for 10 minutes, sketch it on paper, and it clicks.
- Bad practice: Copying code without understanding. You fail the peer defense because you cannot answer conceptual questions.

— *Learning is not knowing the answer; it is developing the ability to find it.* —

Chapter III: Foreword

In 1989, Tim Berners-Lee invented the World Wide Web. At the core of his invention was HTML: a simple, semantic markup language that allows documents to link to one another across global networks.

HTML is not a programming language—it is the skeleton of the internet. It provides structure, accessibility, and meaning to content.

— Every web application starts with an HTML skeleton. —

Chapter IV — Exercise 00: Empty Static Page

Directory: ex00/

Files to Submit: index00.html

Authorized: HTML5

The simplest valid HTML5 page.

- Create a valid HTML5 file named index00.html.
- Must include DOCTYPE declaration, html tag with lang attribute, head, meta charset UTF-8, responsive viewport, and body.
- The page title in the head must be exactly:

%> 42

! The page title must be 42, nothing more, nothing less.

— Structure before style. —

Chapter V — Exercise 01: Text Hierarchy

Directory: ex01/

Files to Submit: index01.html

Authorized: h1-h6, p, hr, br, strong, em

Structuring written knowledge.

- Create index01.html discussing "Why I Am Learning to Code".
- Implement a strict semantic text hierarchy: one h1, at least two h2 subheadings, h3, multiple paragraphs (p), bold importance (strong), and emphasis (em).
- Use hr to partition thematic sections and br for line breaks.

! Never skip heading levels (e.g. going from h1 directly to h4).

— *Accessibility begins with semantic text.* —

Chapter VI — Exercise 02: Links & Images

Directory: ex02/

Files to Submit: index02.html, assets/profile.jpg

Authorized: a, img, relative/absolute URLs

Connecting resources and visual media.

- Create an assets/ subfolder containing your profile photo.
- In index02.html, embed the photo using a clean relative path (assets/profile.jpg) with a descriptive alt attribute.
- Include an external link to GitHub opening in a new tab with rel="noopener noreferrer".
- Include an internal anchor jump link (#contact) and an email link (mailto:).

— A link without descriptive alt text is invisible to screen readers. —

Chapter VII — Exercise 03: Lists & Tables

Directory: ex03/

Files to Submit: index03.html

Authorized: ul, ol, li, table, thead, tbody, th, td, caption

Tabulating and enumerating data.

- Build an unordered list (ul) of favorite programming languages.
- Build an ordered list (ol) detailing a step-by-step workflow with a nested sub-list.
- Build a semantic data table of a weekly study timetable containing caption, thead, th with scope="col", tbody, and td data cells.

! Never use HTML tables for visual page layout. Tables are strictly for tabular data.

— *Nesting lists requires placing child lists inside li tags.* —

Chapter VIII — Exercise 04: Forms & Inputs

Directory: ex04/

Files to Submit: index04.html

Authorized: form, input, label, textarea, select, button

Collecting user input.

- Build an accessible user registration form inside index04.html.
- Include input controls for text (Full Name), email, password, radio buttons (group), checkbox (agreement), select dropdown, and textarea.
- Pair every input with a matching label element using for="id".
- Apply native HTML5 validation: required and minlength.

— *Clicking a label should always focus its input.* —

Chapter IX — Exercise 05: Semantic Architecture

Directory: ex05/

Files to Submit: index05.html

Authorized: header, nav, main, section, article, aside, footer

Synthesizing the complete web document.

- Synthesize all HTML elements into a unified technical blog page index05.html.
- Structure the document using HTML5 landmarks: header, nav, main, section, article, aside, and footer.

! Minimize generic div usage; use semantic landmarks to give structure to your content.

— *Semantic HTML makes the web open to all humans and search engines.* —

Chapter XIV: Submission & Evaluation

- Submit your assignment to your Git repository as instructed. Only the work pushed to your repository branch will be evaluated during defense.
- Double-check your directory structure, file permissions, and exact file names before final submission.

! You must submit only the files explicitly required by the project instructions.

- Prepare to demonstrate your code live, explain your logic line-by-line, and answer questions about core concepts.

— *RTFM and verify your work before peer defense!* —